

CHINA LAKE, CA, USA

WMO: 746120

Lat: 35.688N

Lon: 117.693W

Elev: 680

StdP: 93.42

Time Zone: -8.00 (NAP)

Period: 07-19

WBAN: 93104

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
12	-4.9	-3.1	-22.0	0.6	8.5	-19.4	0.7	8.5	14.0	12.9	12.4	12.8	1.3	200	0.519

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
7	17.5	42.4	18.4	41.1	18.1	39.9	17.6	20.5	36.4	19.8	36.4	19.1	36.1	4.8	170

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
15.3	11.8	26.3	13.3	10.3	28.5	11.4	9.1	27.5	61.9	36.3	59.2	36.4	56.6	36.3	23.5

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
			Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
1%	2.5%	5%	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max		
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(4) 12.3	10.9	9.2	DB	-8.4	44.6	1.9	1.0	-9.8	45.3	-10.9	45.9	-12.0	46.5	-13.4	47.2	(4)
(5)			WB	-10.3	22.1	1.8	1.2	-11.6	22.9	-12.7	23.6	-13.8	24.3	-15.1	25.2	(5)

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec		
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(6) Temperatures, Degree-Days and Degree-Hours	DBAvg	18.7	7.7	9.9	14.2	17.8	22.0	27.8	31.2	30.2	25.9	18.6	12.1	6.8	(6)	
	DBStd	9.03	3.29	3.46	3.59	3.77	3.92	3.48	2.47	2.28	3.55	3.46	3.60	3.31	(7)	
	HDD10.0	263	84	42	7	1	0	0	0	0	0	1	19	109	(8)	
	HDD18.3	1351	329	237	133	53	13	0	0	0	2	37	188	359	(9)	
	CDD10.0	3450	13	38	138	236	372	534	658	626	476	268	82	9	(10)	
	CDD18.3	1496	0	0	6	38	127	284	399	368	228	46	1	0	(11)	
	CDH23.3	24014	3	18	213	693	1963	4505	6501	5797	3362	862	98	0	(12)	
	CDH26.7	14832	1	0	47	254	1021	2896	4407	3842	2022	332	11	0	(13)	
(14) Wind	WSAvg	3.4	2.3	3.2	3.9	4.4	4.2	3.9	3.6	3.5	3.2	3.1	2.5	2.6	(14)	
(15) Precipitation	PrecAvg	101	21	27	12	3	2	1	2	2	4	5	7	15	(15)	
	PrecMax	240	130	111	66	36	20	5	18	15	53	30	34	77	(16)	
	PrecMin	21	0	0	0	0	0	0	0	0	0	0	0	0	(17)	
	PrecStd	60	29	29	16	7	4	1	3	3	10	8	8	18	(18)	
(19) Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	22.1	25.5	30.6	34.1	38.4	43.6	44.3	42.7	40.5	34.4	28.1	20.6	(19)	
		MCWB	10.0	10.9	12.8	14.6	15.9	18.0	19.0	18.6	18.5	15.5	12.0	10.0	(20)	
	2%	DB	19.7	23.6	27.9	31.7	36.1	41.4	42.9	41.5	38.7	32.2	26.3	18.4	(21)	
		MCWB	8.4	10.3	12.0	13.2	15.0	17.4	18.7	18.4	17.3	14.9	11.1	8.6	(22)	
	5%	DB	17.8	21.4	25.6	29.7	34.4	39.9	41.6	40.4	37.3	30.3	24.0	16.5	(23)	
		MCWB	7.8	9.5	11.1	12.6	14.4	16.7	18.3	18.0	16.9	13.5	10.7	7.7	(24)	
	10%	DB	15.8	19.0	23.7	27.6	32.4	38.2	40.1	39.2	35.9	28.2	21.2	14.6	(25)	
		MCWB	7.2	8.4	10.4	12.0	14.0	16.3	17.9	17.4	16.2	12.6	9.4	6.9	(26)	
(27) Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	11.9	12.6	13.6	15.3	16.5	19.7	21.7	21.5	20.4	17.9	13.0	11.9	(27)	
		MCDB	17.0	19.2	25.0	30.8	36.2	40.1	33.5	36.5	35.0	28.3	24.7	16.9	(28)	
	2%	WB	10.5	10.8	12.7	14.1	15.6	18.6	20.7	20.5	19.5	16.3	12.0	10.6	(29)	
		MCDB	15.3	21.5	24.4	29.5	33.9	38.1	36.7	36.2	35.3	27.7	23.6	16.0	(30)	
	5%	WB	9.4	10.0	11.9	13.2	15.0	17.7	20.1	19.5	18.4	15.0	11.1	9.2	(31)	
		MCDB	14.7	20.1	24.2	27.7	32.5	37.1	36.5	36.0	33.5	27.2	22.1	14.2	(32)	
	10%	WB	8.2	9.1	10.9	12.4	14.3	16.9	19.4	18.7	17.2	13.9	10.2	7.7	(33)	
		MCDB	14.7	18.0	22.4	26.2	31.0	36.0	36.3	36.0	33.0	25.7	20.3	13.1	(34)	
(35) Mean Daily Temperature Range	5% DB	MDBR	15.6	16.0	16.3	16.6	17.1	18.4	17.5	18.2	18.2	17.8	17.4	14.8	(35)	
		MCDBR	18.7	19.9	19.9	19.5	19.9	20.4	18.8	18.9	19.4	20.4	20.4	17.8	(36)	
	5% WB	MCWBR	10.3	9.7	8.8	7.9	7.2	6.6	6.1	6.5	7.1	9.0	10.4	10.3	(37)	
		MCWBR	13.7	17.9	18.0	17.9	19.2	19.1	15.3	16.3	16.5	16.4	18.3	13.4	(38)	
(40) Clear-Sky Solar Irradiance	taub		0.267	0.277	0.285	0.310	0.327	0.322	0.345	0.334	0.314	0.292	0.269	0.268	(40)	
		taud	2.582	2.570	2.539	2.463	2.440	2.460	2.451	2.455	2.516	2.571	2.604	2.594	(41)	
	Ebn at Noon		952	978	991	974	954	954	930	937	945	946	943	930	(42)	
		Edh at Noon	75	85	96	109	114	112	112	109	97	84	72	70	(43)	
(44) All-Sky Solar Radiation	RadAvg		3.03	4.03	5.52	6.91	7.85	8.55	7.95	7.43	6.34	4.75	3.49	2.75	(44)	
	RadStd		0.23	0.37	0.24	0.25	0.32	0.30	0.34	0.22	0.17	0.24	0.14	0.20	(45)	

Historical Trends

		DBAvg	Heating		Cooling			Degree-Days					
			99% DB	99% DP	1% DB	1% WB	1% DP	HDD10.0	HDD18.3	CDD10.0	CDD18.3		
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)		
(46) Station Only		N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	(46)	
(47) Regional (28 neighbors)		+0.64	N/A	-1.70	+0.43	N/A	N/A	N/A	-116	+212	+115	(47)	

Nomenclature: See separate page